

Using simulation modeling to inform management of invasive species: A case study of eastern brook trout suppression and eradication

Casey C. Day^{a,*}, Erin L. Landguth^b, Andrew Bearlin^c, Zachary A. Holden^d, Andrew R. Whiteley^e

^a Department of Forestry and Natural Resources, Purdue University, 195 Marsteller Street, West Lafayette, IN 47909, USA

^b School of Public and Community Health Sciences, University of Montana, 32 Campus Drive, Missoula, MT 59812, USA

^c Seattle City Light, Environment, Land and Licensing, 700 5th Ave, Seattle, WA 98124, USA

^d USDA Forest Service Region 1, Missoula, MT 59804, USA

^e Franke College of Forestry and Conservation, Wildlife Biology Program, University of Montana, Missoula, MT 59812, USA

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ABSTRACT

Ecosystem impacts due to invasive species continue to attract significant conservation effort worldwide. In aquatic ecosystems, physical interventions such as suppression and eradication of non-native species are typically expensive, long-term commitments, with few examples of lasting success in the absence of significant ongoing effort. Control of non-native species is a major conservation and restoration challenge, as a species' demographic resilience and connectivity within networks can limit the ability of suppression or eradication efforts to influence populations. Simulation tools can provide valuable insights for the management of these systems - from evaluation of tradeoffs between time and effort to prediction of relative success rates of alternative strategies in changing environments. In the Pacific Northwest region of the U.S., the eastern brook trout (EBT; *Salvelinus fontinalis*) is a non-native invasive species that competes with native fish species across a wide spatial scale due to extensive human-mediated introduction starting in the early 20th century. The goal of this study was to simulate the individual movement and demographics of EBT before, during, and following implementation of control efforts in tributaries within the Pend Oreille River watershed. The ultimate purpose of the model was to inform mitigation decisions through the investigation of alternative management actions in an adaptive management framework. Our results indicate that eradication of EBT is improbable in large systems via electrofishing, but suppression is a viable alternative given sustained management efforts. Changes to scheduling, effort, and length of electrofishing suppression treatments had minimal effects on EBT population recovery times. We reproduced the effects of compensatory responses to control treatments, including increases in juvenile survival and emigration rates, and demonstrated that these mechanisms are likely drivers of recovery following treatment. Our study highlights the many benefits of incorporating spatially explicit, individual-based models into management plans for the control of invasive species.

1. Introduction

Invasive species are increasingly threatening native ecosystems. In the United States, damages from these species have been estimated to cost between 121 and 220 billion USD annually (Pimentel et al., 2005; Marbuah et al., 2014), and have been identified as a major cause of past species extinctions (Clavero and García-Berthou, 2005). Invasive species can negatively impact ecosystem services that are beneficial to industries such as agriculture, water, tourism, and recreation (Charles and Dukes, 2008). In response to the damage caused, tremendous effort is spent on preventing the establishment of new invasions, as well as mitigating existing invasions, though results of such programs are

rarely documented (Marbuah et al., 2014). Approaches to control of invasive species range from prevention via economic incentives, policy instruments, and preventive monitoring, to on-the-ground management and in some cases, direct action against the invasive species via various methods (Genovesi and Shine, 2004; e.g., mechanical harvest, biological control, genetic manipulation, and systematic eradication).

Eradication (i.e., complete removal of a species) and suppression (i.e., reduction of species abundance/density) are approaches commonly taken to control invasive species. Some eradication programs have been successful as eradications of diverse taxa have been reported across the globe (e.g., rodent eradications from islands, Howald et al., 2007; plant eradications resulting in restored ecological function,

* Corresponding author.

E-mail address: caseycday@gmail.com (C.C. Day).

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Kennedy et al., 2005; Holsman et al., 2010; and insect eradications, Klassen and Curtis, 2005). A review of attempted eradications in Europe reported 37 successful eradications of mammals, but noted limited success among plant and invertebrate species (Genovesi, 2005). Other programs may fail at eradication but are successful at suppressing abundance of invasive species to low densities. The ecological consequences of suppression and whether it is functionally equivalent to eradication remains unclear (Simberloff, 2009). Given the environmental, economic, and social ramifications of eradication and suppression programs, it is vital to strike a balance between effort, financial cost, ecosystem integrity, monitoring, and prevention (Fraser et al., 2006; Mehta et al., 2007; Bogich et al., 2008; Lampert et al., 2014).

Ecological modeling is one approach to the management of invasive species that can be implemented at relatively low cost in terms of money, effort, and ecosystem function. Models can be developed and evaluated prior to and after manipulation of the natural system. Examples include models that have evaluated the cost of eradication versus suppression (Baxter et al., 2008), predicted probable invasion locations (Schmidt et al., 2010), and examined risks posed by potential invaders through distributional niche modeling (Jiménez-Valverde et al., 2011). Spatially-explicit individual-based models (SIBM) are a type of ecological model becoming more common as tools used for understanding mechanisms driving invasion dynamics (Pyšek and Hulme, 2005; Anderson and Dragičević, 2015; Fraser et al., 2015). A spatially-explicit model can allow for landscape configuration to affect model processes, and thus, for testing of alternative landscapes under different scenarios of management or land use. A model that is individual-based produces broad-scale patterns that emerge from the interactions of individual agents; a powerful approach to understanding mechanistic drivers of population-level processes (Grimm and Railsback, 2005). Invasions are inherently population-level processes (Peterson et al., 2004a), and therefore, modeling those processes from the bottom-up may reveal insights as to how invaders are successful and what strategies may succeed in the management of invasive species.

We developed a SIBM to understand the processes driving the persistence of eastern brook trout (hereafter EBT; *Salvelinus fontinalis*) invasions of streams in the northwestern United States, and to evaluate alternative management strategies for the control of EBT. EBT are native to the eastern United States, but have been repeatedly and intentionally introduced to western stream systems for recreational purposes (Fuller et al., 1999; Dunham et al., 2002). These introductions, and subsequent colonizations of contiguous habitat have been identified as a substantial threat to the persistence or recovery of native salmonids through interbreeding or competition for habitat and food resources (Andonaegui, 2003). In the Pacific Northwest, EBT are often implicated for having negative effects on native species such as Westslope Cutthroat Trout (*Oncorhynchus clarkii lewisi*; Dunham et al., 2002; Scoppettone et al., 2012), Bull Trout (*Salvelinus confluentus*; Gunkel et al., 2002) and Redband Rainbow Trout (*Oncorhynchus mykiss gairdnerii*; Miller et al., 2014).

Eradication and suppression programs are a common management tool used in response to EBT invasion and establishment, and are often undertaken with the goal of restoring native trout populations. Results from such programs have been mixed. Eradication has occasionally been successful in smaller systems, but in larger systems suppression may serve as a manageable substitute to aid in the restoration of native species (Shepard et al., 2002; Meyer et al., 2006; Carmona-Catot et al., 2010). One factor affecting the success of these programs is compensatory responses, or the ability of EBT to compensate for the additional mortality of control treatments via increased survival or immigration rates (Peterson et al., 2008). SIBMs are an ideal tool for evaluating eradication and suppression programs, investigating compensatory mechanisms, and preventing the loss of money and time due to failed efforts. Such models can then be implemented in an adaptive management framework, adjusting the model based on feedback from empirical data collected during field operations, with concomitant

improvement of applied management actions over time (Starfield and Bleloch, 1986; Walters, 1986).

The broad objective of our simulation study was to produce a model that would simulate the bottom-up mechanisms driving EBT demographics (e.g., movement, mortality, growth) that could then be used to answer questions about how EBT respond to control treatments. Specific objectives were to (1) evaluate and compare alternative management strategies for EBT control, and (2) investigate the role of compensatory responses to control efforts by EBT. We examined the effect of the following factors on EBT suppression, recovery (i.e., return of population size to pre-suppression levels), and compensatory responses: suppression schedule, effort, landscape configuration (i.e., stream barriers), and stray rates (i.e., dispersal rates). Our results will be used to inform an active adaptive EBT management program taking place in the study area. Our results and methodology will also serve as a template for simulating the management of invasive aquatic species, particularly in the context of conservation of threatened species and ecosystems. Finally, our results demonstrate the value of applying SIBMs to landscape-level problems and the ability to test alternative mechanisms driving system processes.

2. Methods

2.1. Study system

We modeled the manual and chemical control of non-native EBT in north-eastern Washington, USA (Fig. 1) in the lower Pend Oreille River and nearby tributaries. The study area on the lower Pend Oreille River extends from Box Canyon Dam (RM 34.5) to Boundary Dam (RM 16.0) located on the US-Canada border. Major tributaries on this river stretch include Flume, Slate, Sullivan, and Sweet Creeks. The Pend Oreille River originates in Montana and Idaho, USA, and flows through the Okanogan Highlands in Washington, USA, the western portion of the Rocky Mountains, and enters the Columbia River a few miles north of the border between Washington and British Columbia, CA. Where native fish once moved freely, dams, culverts, and warm impounded waters have restricted fish travel or allowed only one-way passage out of tributaries. Stocked non-native fish (e.g., EBT) have interacted both ecologically (e.g., competition of resources) and biologically (e.g., interbreeding) with their native counterparts. EBT were cultivated and stocked widely throughout the Western US during the 20th century (Fuller et al., 1999), primarily from state operated hatcheries to promote fishing opportunities for recreation and subsistence. Within the region of interest to this study approximately 665,000 EBT were stocked into many local tributaries over the period of record from 1933 to 1981 (Kloempken, 1996) leading to extensive naturalization, range expansion and attendant impacts on native fauna. EBT currently tend to occupy the lower reaches of tributaries where they are sympatric with native Westslope Cutthroat trout and Bull Trout (*Salvelinus confluentus*) that are listed under the Endangered Species Act. Also within the study area are two hydropower dams regulated by the Federal Energy Regulatory Commission that have recently received new operating licenses containing requirements to develop and implement management plans to address the impacts of continued presence of EBT (FERC, 2005, 2013).

2.2. Model overview

We used CDMetaPOP v1.0 (for a full model description see Landguth et al., 2017) to simulate the proposed EBT management plan (SCL, 2010). CDMetaPOP is a riverscape-level, spatially-explicit, and individual-based eco-genetic model of meta-population dynamics. CDMetaPOP simulates demographic (and genetic) processes of a meta-population comprising a number of populations inhabiting individual patches. Patches are populated with individuals and regulate carrying capacity, habitat suitability, temperature, capture probability, and

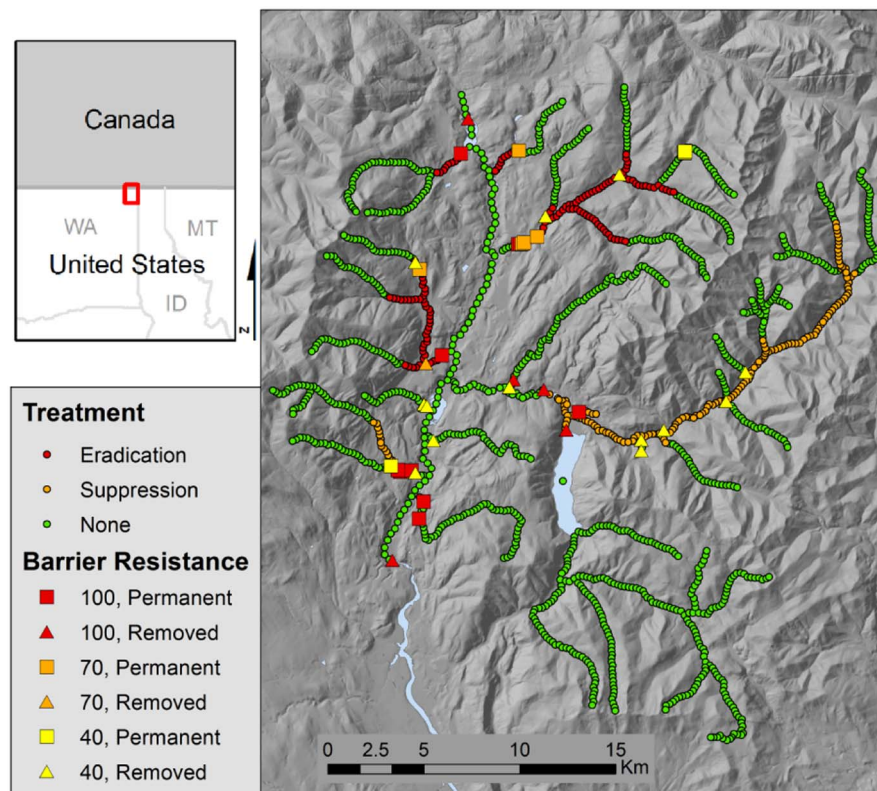


Fig. 1. EBT modeling extent with treatment scenarios represented by color and barrier types represented by shape. Those barriers labeled for removal were scheduled by local management agencies to be removed according to the watershed management plan (SCL, 2010).

resistance to movement between patches. Individuals in the model can mature, mate, produce offspring, grow, migrate, disperse, and die.

In a review of studies on EBT invasions, Dunham et al. (2002) called for the development of a predictive understanding of invasions to improve the prevention and management of EBT populations. We selected CDMetaPOP to simulate control of EBT populations because it uses bottom-up, mechanistic processes to produce patterns of behavior and demographics at scales large enough to simulate our entire study area. This is accomplished by using individual fish as the primary agent in the model, parameterized with simple rules to interact with the environment that collectively result in population-level patterns. In this way, a model that produces realistic patterns can then be applied to other contexts or to make predictions about population-level responses to changes in environmental conditions or management strategies. In addition, CDMetaPOP is spatially explicit, and uses user-defined resistance surfaces to regulate movement as opposed to simple dispersal kernels. Again, simple rules assigned to individual patches interact with individuals to produce population-level patterns of movement and demographics. CDMetaPOP also supports asymmetrical landscape resistance, a vital component when simulating barriers to fish movement such as dams or culverts. Following, we introduce CDMetaPOP processes and parameters as they relate to our EBT system model. For an overview of main processes and scheduling, see Fig. 2. For more detailed information on all the processes simulated in CDMetaPOP, see the user manual (<https://github.com/ComputationalEcologyLab/CDMetaPOP/>).

2.3. Patches

The EBT simulation extent contained 1384 patches, or spatially delineated stream reaches of ~200 m each (Fig. 1). Initial presence or absence of EBT and habitat suitability of a given patch was determined based on field sampling data (West Fork Environmental, 2012). We simulated occupancy and abundance of EBT in each patch during two

distinct seasons: spawning and overwintering.

Carrying capacity for each patch was calculated based on electrofishing sampling efforts and multiple-pass depletion surveys (Peterson et al., 2004b; Bearlin and Simmons, 2015). A regression model was used to model electrofishing capture probability (i.e., probability of mortality) by age and fish abundance as a function of slope and aspect in the riverscape system.

To determine temperature values for each patch, we developed a daily 30 m resolution gridded map of stream temperatures using data from a network of 34 HOBO Tidbit and ProV2 temperature sensors (Onset Computer Corp., Bourne, MA, USA), distributed across the EBT model study area from July 1st 2013–July 23rd 2015. Sensors were installed following Isaak et al. (2013) and were shielded from direct solar exposure and placed in deep locations to avoid dewatering and icing events. We used Principal components analysis (PCA) to identify the modes of spatial and temporal variation in the stream temperature data. For each site, daily average stream temperature time series were first calculated from the raw 4-hourly observations.

The first principal component captured 98% of the total variation common among sensors. The dominant principal component score represented the mean temperature across all sites, and is well correlated with measured temperatures at the Boundary Dam ($r = 0.92$). The principal component loadings represented the spatial variation among sites. We developed a regression model for the first principal component loading (PC1) using flow accumulation (ESRI ArcGIS Desktop v. 10.2, Redlands, CA) and elevation derived from 30 m National Elevation Data (NED). The resulting model explained 81% of the variation in PC1 and was used to estimate the PC1 loading at each 30 m grid cell in the stream network. Then, following Holden et al. (2011) we reconstructed stream temperatures at each grid cell by combining the modeled loadings and observed principal component score, resulting in a daily predicted stream temperature map at 30 m resolution raster grids. Overall, the model performed well at most sites, with mean absolute errors of less than 1 °C. We extracted modeled average

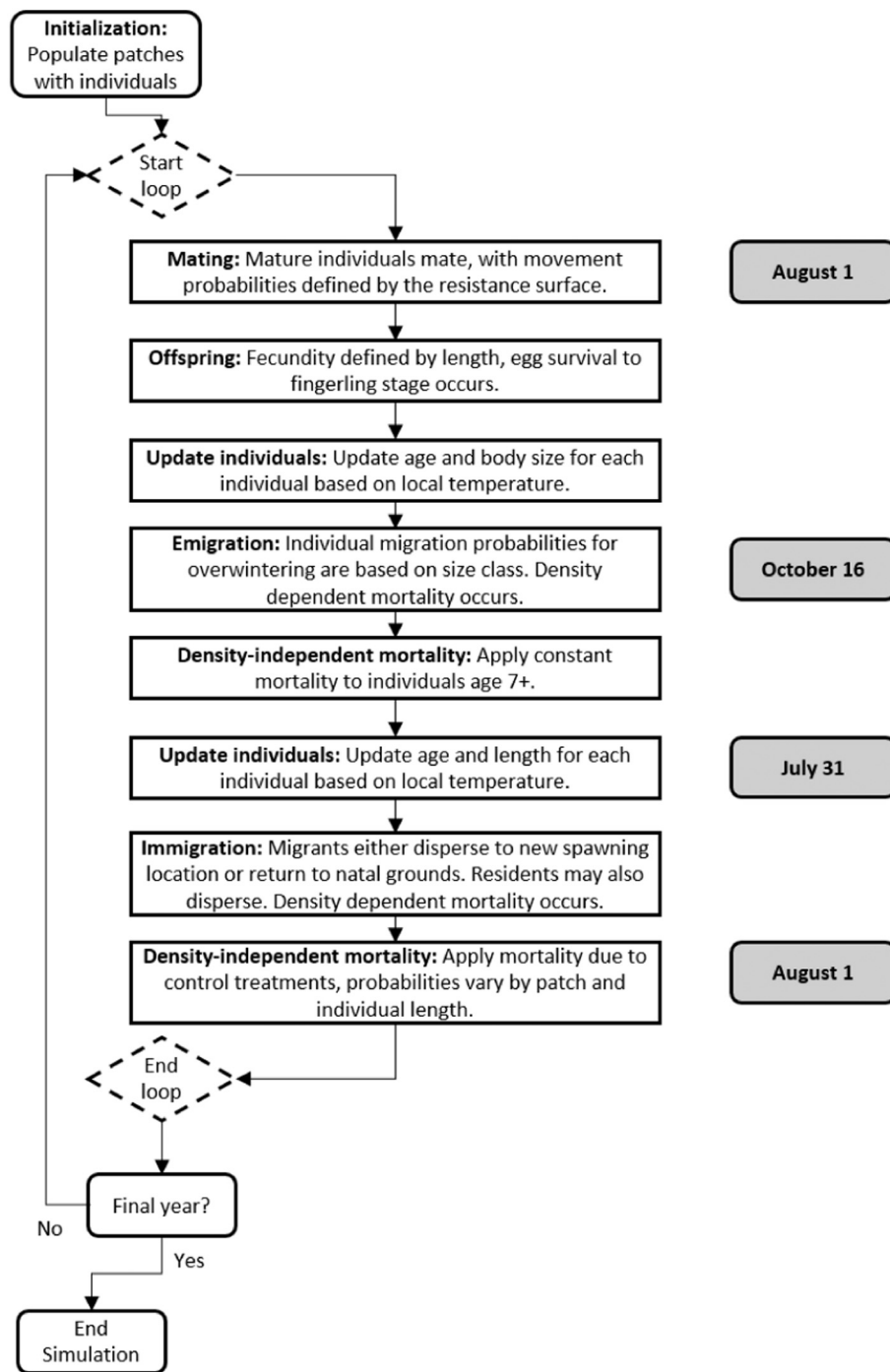


Fig. 2. Overview of model processes and scheduling implemented for simulations of eastern brook trout control treatments in northeastern Washington, USA.

temperatures separately for the spawning period (Aug. 1st–Oct. 15th) and the overwintering period (Oct. 16th–July 31st).

2.4. Individuals

Probability of maturation for individual fish was modeled as a continuous function of size and calculated each year based on a logistic probability curve (e.g., Downs et al., 1997) derived from maturation data provided by SCL (Fig. A2). In order to provide an over-riding control on maturation of slow-growing animals, individuals became mature at age 6 regardless of their size.

Fecundity was modeled as a linear function of size (e.g., Downs et al., 1997). This model was fit using data collected by SCL (West Fork

Environmental, 2012).

$$\text{Number of eggs} = 3.78 * \text{Length (mm)} - 445. \quad (1)$$

Then, mean number of eggs for each spawning occurrence was drawn based on a Poisson deviate. Mature individuals in the model were capable of reproducing every year, and females were limited to one egg-laying event per year.

Growth was based on the Von Bertalanffy equation (von Bertalanffy, 1938) and adjusted based on spatially-explicit temperature values. For simplification, we assumed that temperature was the primary environmental driver of growth in the system rather than attempt to reproduce the interactive effects of temperature, flow, density, and food availability (Jobling, 1997; Xu et al., 2010). For our EBT model, we

Table 1

Summary of the factors that were varied and their levels during simulation of electrofishing suppression of eastern brook trout in northeastern Washington State, USA.

Factor	Levels	Values	Notes
Number of 10-year suppression cycles	2	1, 3	3-cycle runs were conducted using only landscapes IBD and FPB and at standard level of effort.
Electrofishing effort	4	None, low, standard, high	High and Low effort scenarios were conducted using only landscapes IBD and FPB, 1 suppression cycle, and electro-fishing schedule 2.
Landscape ^a	5	IBD, ECB, FCB, EPB, FPB	
Stray rate	3	0, 0.001, 0.01	Values indicate probability of straying per individual per generation.
Electrofishing schedule (electrofishing passes per year)	4	No electrofishing 1 – 2,2,1,1,1,1,1,1,1,1 2 – 3,0,0,3,0,0,3,0,0,3 3 – 3,3,3,3,0,0,0,0,0,0	Electrofishing schedules were staggered by 2 years between the upper and lower sections of Sullivan Creek.

^a IBD = isolation by distance; ECB = existing complete barriers; FCB = future complete barriers; EPB = existing partial barriers; FPB = future partial barriers.

used the following equation and parameters derived from data provided by SCL (West Fork Environmental, 2012) to produce incremental length (mm):

$$L_{\text{inc}} = L' \exp(k' * (i + 1)) * (GD/365), \quad (2)$$

where L_{inc} represents the seasonal growth increment, L' represents the new length from the original Von Bertalanffy equation, k' represents the maximum growth rate adjusted for observed temperatures, and GD represents the number of grow days in the season. For a full derivation and examples, see the CDMetaPOP user manual section 1.3.1.7.2.

Several parameters were governed by size class, rather than the size of each individual. We simulated eight size classes based on size distributions from electrofishing data obtained from Seattle City Light (SCL, 2010; West Fork Environmental, 2012). Size classes (mm) were 60 (fingerlings), 110, 140, 180, 200, 230, 240, and > 250. Size was initialized using a normal draw for each respective age class around empirically derived standard deviations (3, 11, 17, 18, 17, 19, 20, and 20 mm, respectively). Parameters associated with size class (i.e., probability of mortality, migration, and dispersal) varied as fish grew and entered subsequent size classes.

In our application of CDMetaPOP, mortality occurred as both a density dependent and density independent process. Density dependent mortality occurred after emigration to overwintering patches and after immigration or dispersal to spawning grounds. This mortality was regulated by a size-structured carrying capacity that represented intra-specific competition for limited resources (e.g., Ray and Hastings, 1996; see CDMetaPOP user manual section 1.3.1.4.4 for more details on this algorithm). Density-independent mortality that varied by patch and size class occurred following immigration or dispersal, and was used to implement alternative management scenarios of suppression and eradication (see Study design section). To reduce the rare occurrences of long-lived fish in the system model, an additional density-independent 50% probability of mortality was implemented for fish age seven and above. We excluded explicit mortality due to inter-specific competition, and assumed that the occurrence of Cutthroat Trout did not have a negative effect on EBT populations (Peterson et al., 2004a; McGrath and Lewis, 2007).

Our model considered four movement processes: mate-searching, migration-out (away from natal grounds), migration-back, and straying. Probability of migration-out varied with size class, and each individual had some chance of migrating out of the natal patch each year. For each of the eight size classes, probability of migration was 0.0, 0.1, 0.2, 0.3, 0.4, 0.4, 0.4, and 0.4, respectively. These migration probabilities are unknown in this particular system. However, preliminary sensitivity analysis conducted around these values demonstrated minimal variation in demographic response variables. In the simulation, migration took place as two separate events. If assigned to migrate as a function of the size-specific migration probability, individuals moved to their assigned patch to overwinter. At the end of the overwintering period and just prior to spawning season, all individuals had an equal probability

of straying to a new location, depending on the simulation scenario. Migrants not assigned to stray returned to their natal spawning patch.

2.5. Movement

Riverscape structure governed movement of individuals between patches and this concept was implemented through resistance surfaces (described below) whose values represent the stepwise cost of moving between two adjacent raster cells. (e.g., Spear et al., 2015). For each riverscape scenario, we used UNICOR (Landguth et al., 2012) to create a 1384×1384 probability matrix based on the resistance surface and the cumulative cost between each pair of patches. Movement between patches was then simulated as a probabilistic function based on this probability matrix (e.g., Landguth and Cushman, 2010). For scenarios where barriers occurred between two patches, we multiplied the movement probabilities between patches by the permeability value of each barrier (i.e., 0.0, 0.3, or 0.6). We also incorporated thresholds to limit the maximum distance available to a fish for each movement process. The threshold for local mating movement was 2 km (Gowan and Fausch, 1996; Peterson and Fausch, 2003; Petty et al., 2012), and the threshold for migration and straying movement was 25% of the maximum cost distance between any two patches on the landscape (~15 km in an IBD scenario; Shetter, 1968; Adams et al., 2001; Dunham et al., 2002).

2.6. Study design

We modeled the control of EBT in north-eastern Washington in accordance with a management plan developed by Seattle City Light (SCL, 2010; Fig. 1). The plan consisted of both eradication via chemical treatment (i.e., rotenone) in some streams and various patterns of electrofishing schedules for the suppression streams (see Table 1 and Table A2 for a full list of model scenarios). All management options (and therefore, all simulations) contained both eradication and suppression treatments. Invasive species control measures were implemented in the simulations through both patch- and sized-based mortality. Patches designated for chemical eradication underwent 100% mortality for that year, and were subject to immigration in subsequent years. To simulate suppression via electrofishing, individuals occupying suppressed patches underwent a size-dependent probability of mortality that was multiplied by a patch-specific capture probability derived from SCL capture data. The base size-dependent probabilities of mortality per size class were 0.1, 0.45, 0.75, 0.95, 1, 1, 1, and 1, respectively. These probabilities were then multiplied by patch-based capture probabilities ranging from 0.39–0.58. We then adjusted the resulting mortality probabilities according to the number of planned electrofishing passes per year to produce final mortality rates. Patches that were neither suppressed nor eradicated in a given year experienced no density-independent mortality.

We evaluated the response of EBT to control treatments while

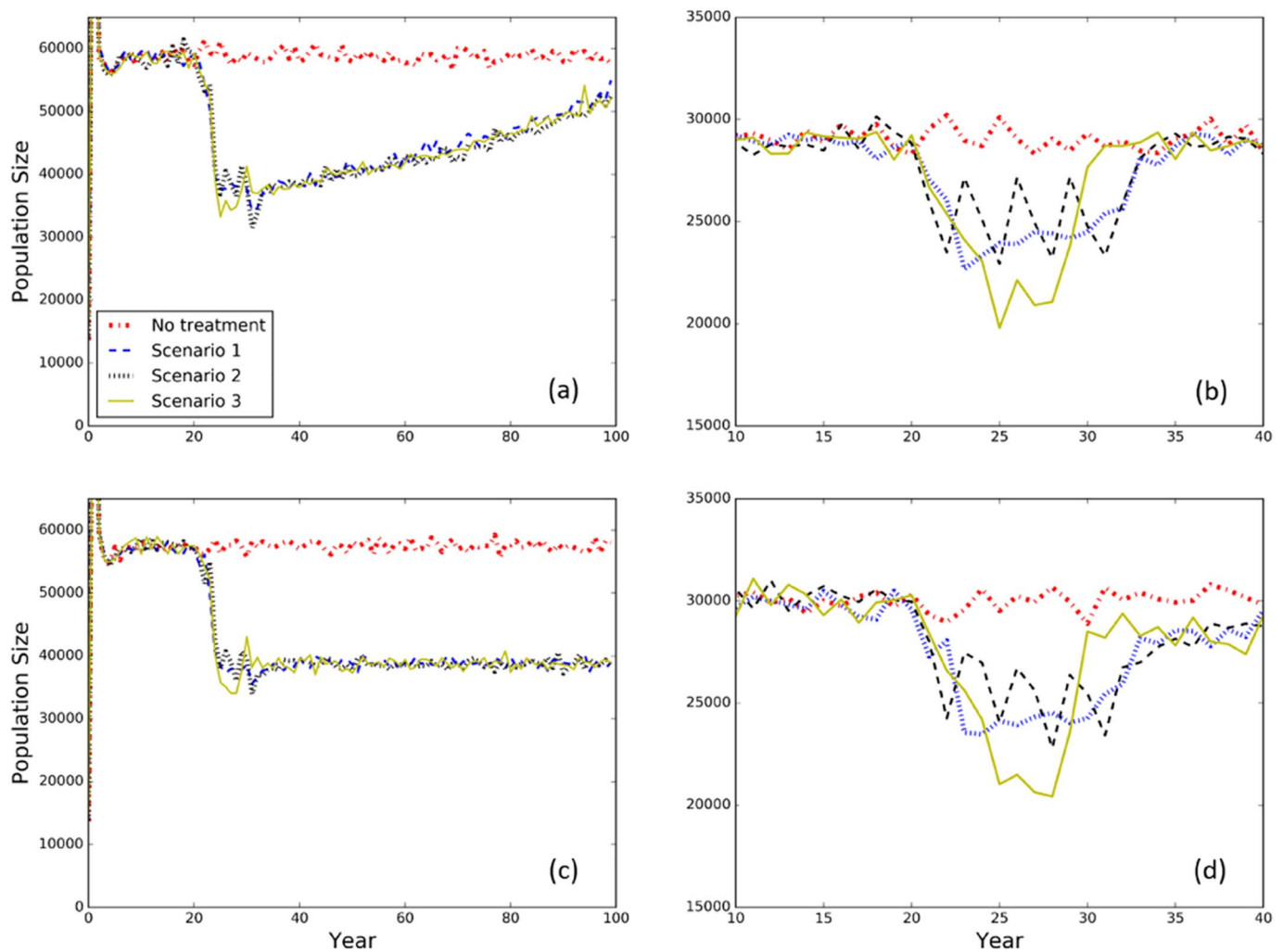


Fig. 3. Simulated population levels of EBT in the Lower Pend Oreille River, Washington, USA, with alternative strategies of suppression and eradication activities beginning at year 20. The top panels display the isolation-by-distance riverscape, wherein no dams or barriers were considered to impact movement. Panel (a) represents the total landscape population, while (b) represents the subset of patches that received suppression treatment. The bottom panels display results of simulations run on a riverscape surface incorporating both complete and partial barriers to fish movement in the upstream direction. Panel (c) represents the total population, while panel (d) represents the subset of patches that received suppression treatment. Stray rate was 0.001 for all scenarios.

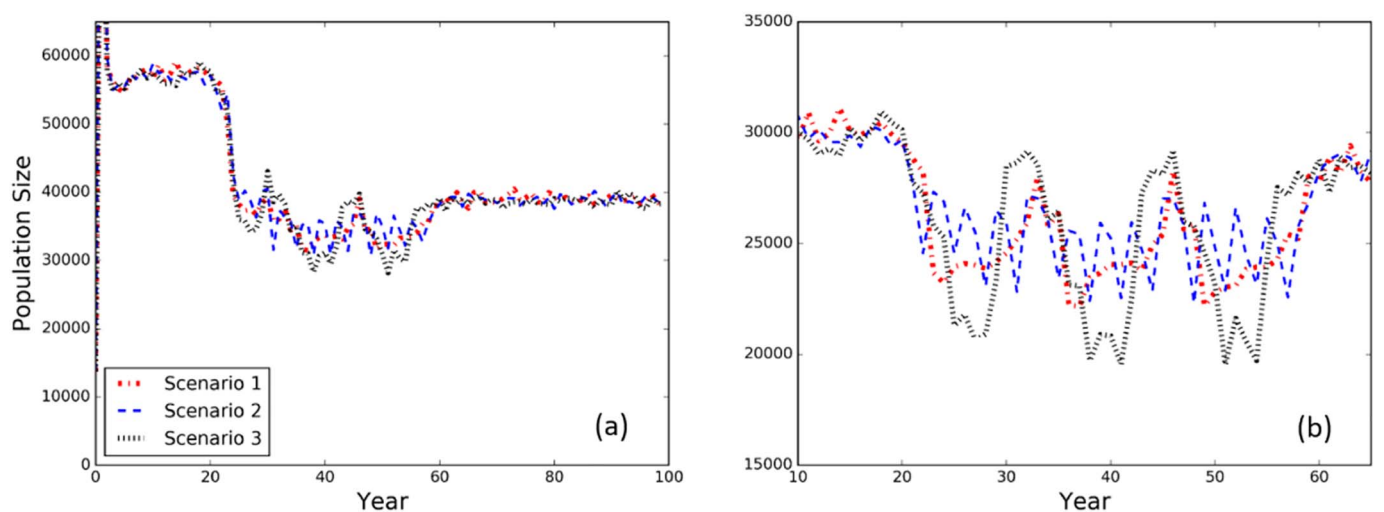


Fig. 4. Three consecutive cycles of EBT suppression displayed for (a) the total population and (b) the subset of patches that received suppression treatment. Note that eradication only occurred in the first cycle.

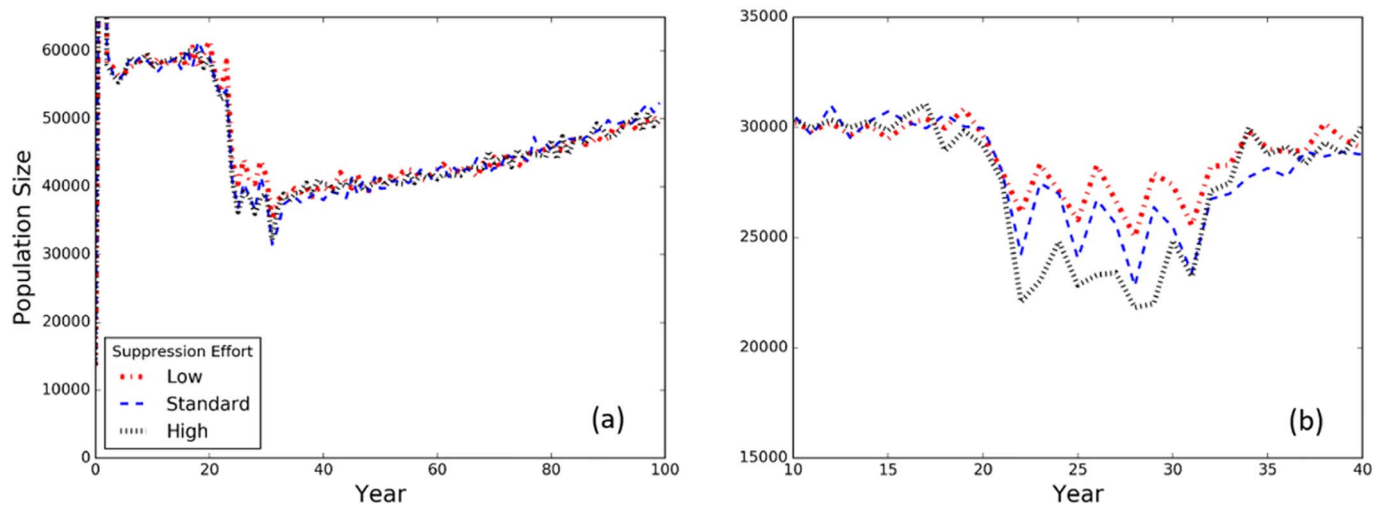


Fig. 5. Alternative levels of EBT suppression effort in an IBD landscape for (a) the total population and (b) the subset of patches that received suppression treatment.

varying 5 factors: suppression schedule, number of cycles of suppression (10 years vs 30 years), effort, stray rates, and landscape configuration (Table 1). Each of these factors and their levels are described in detail below.

(1) We simulated scenarios with no suppression or eradication as a null scenario, and three eradication plus suppression scenarios from a list of possible electrofishing schedules provided by SCL in 10-year cycles with a set number of electrofishing passes per cycle (see Table 1). The spatial arrangement of treatments did not vary across treatment scenarios (see Fig. 1 for a complete display of treatments across patches). In the model, an electrofishing pass was implemented as mortality resulting from traversing a reach of stream from end to end with a backpack or barge electrofisher and euthanizing captured EBT. Each 10-year scenario was allotted equal effort, with 12 total electrofishing passes distributed among years. Scenario 1 spaced passes and effort as evenly as possible across all 10 years. Scenario 2 was staggered effort throughout the suppression cycle, with 2-year periods of no treatment between electrofishing events. Scenario 3 grouped all effort into four straight years of three passes each (Table 1).

(2) Most simulation runs included a single 10-year cycle of electrofishing activities. To test for a cumulative effect of prolonged suppression, we simulated 3 cycles, or 30 years of suppression activities. In

these simulations, eradication was not repeated during the second and third cycles.

(3) We also simulated three levels of suppression effort (low, standard, and high). Standard effort represented the baseline plan of 12 passes per 10 years, low included 8 passes per 10 years (e.g., 4 occasions of 2 passes each), and high included 18 passes per 10 years.

(4) Stray rates are generally unknown for EBT, and so we tested a range of parameter values to determine what effect this rate had on the recovery of suppressed EBT populations (e.g., Peterson et al., 2004a). We tested stray rate values of 0, 0.001, and 0.01.

(5) We developed five primary riverscape surfaces that governed fish movement for mating, migration, and straying events (see Fig. 1). (1) The first riverscape surface served as a null model of movement through the riverscape (i.e., isolation-by-distance; IBD). In IBD, fish had an equal probability of moving to any two suitable locations on the riverscape that were equidistant (riverine distance, not Euclidean) from the fish. In other words, no variables (other than distance and habitat suitability) were considered as contributing to the likelihood of a fish successfully moving to a new patch. The next four riverscape hypotheses all included IBD, but added alternative barrier configurations. (2) IBD + existing complete barriers: In this scenario movement across the landscape was governed by probability values due to distance and

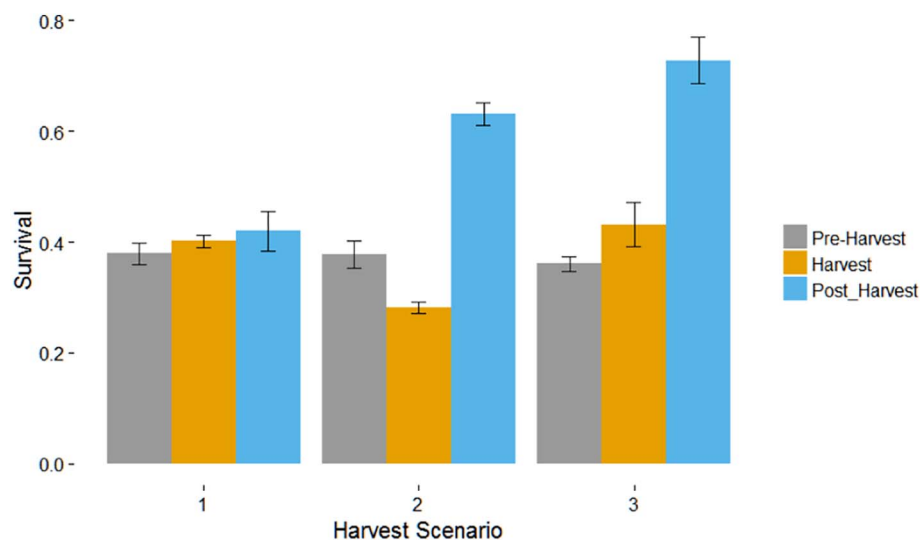


Fig. 6. Patterns of survival of age 0 fish before, during, and after electrofishing suppression of eastern brook trout in northeastern Washington State, USA. Suppression scenarios varied by electrofishing schedule and intensity during those years when suppression was implemented (see Table 1).

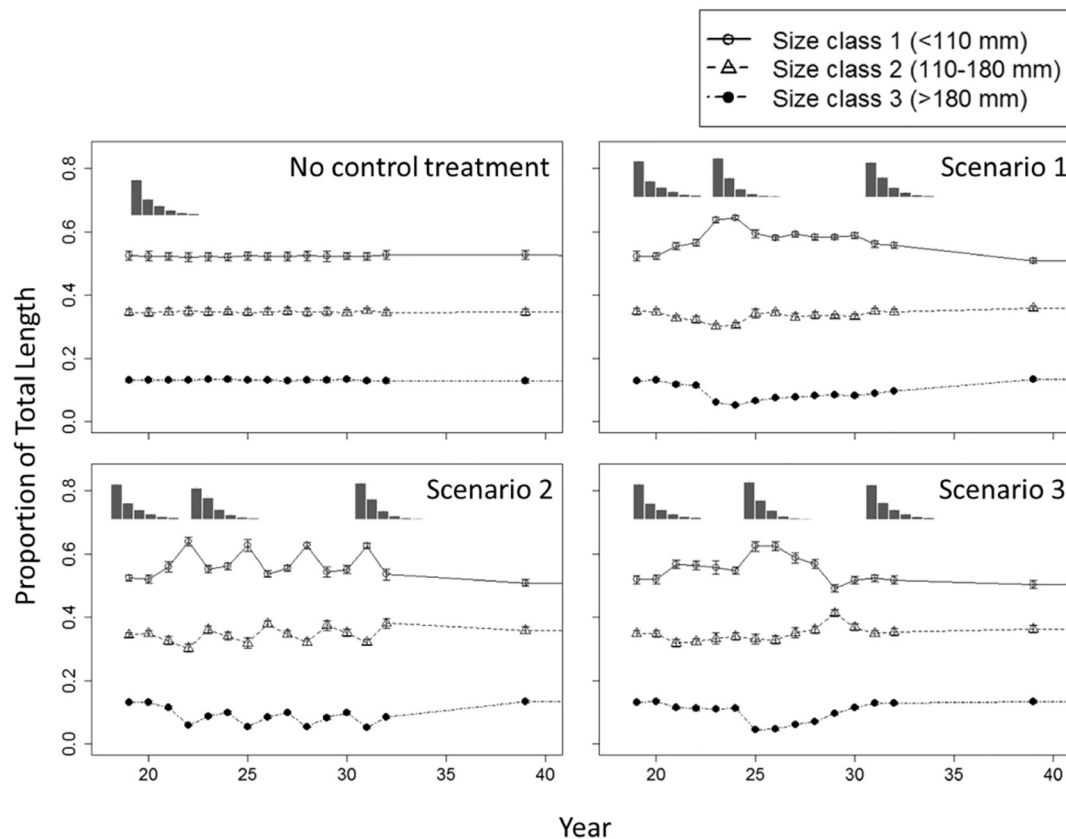


Fig. 7. Size structure present in the subset of patches that underwent suppression treatment for the control of invasive eastern brook trout in northeastern Washington, USA. Control treatments began in year 20, and varied by treatment scenario. Line graphs show the proportion of total fish length represented by the respective size class in a given year across all treated patches, and averaged across 15 replicates of each scenario. Error bars represent one standard deviation. For display purposes, the number of size classes was reduced from 8 to 3. Bar graphs display the proportion of individuals in each of the original 8 size classes, with the 3 largest classes combined to form a single class. In each panel, bar graphs are displayed pre-treatment, during treatment, and in the final year of treatment.

existing complete (e.g., one-way dams, permeability = 0) barriers on the riverscape that prevented movement over the barrier in the upstream direction. Partial barriers (e.g., culverts) were not considered. (3) IBD + future complete barriers: This was identical to scenario 2, but planned barrier removals were simulated according to proposed management plans (SCL, 2010; Fig. 1). (4) IBD + partial existing barriers: This was identical to scenario 2 described above, but with barriers of both complete and partial permeability (e.g., culverts) in the upstream direction included. (5) IBD + partial future barriers: This scenario included future barrier removals and also considered partial barriers in addition to complete barriers.

For each riverscape scenario we created raster surfaces with values between 0 and 1 that represented the permeability of each cell (i.e., 1 = complete passability and 0 = complete impassability). Barrier cells were assigned permeability values relative to the assigned barrier passability (either 0.0, 0.3, or 0.6; Walker et al., 2015), and all other cells were assigned permeability values of 1. We used a riverscape connectivity program, UNICOR (Landguth et al., 2012), to create 1384×1384 probability matrices for each scenario. In these probability matrices, the probability of moving to any given patch was determined by the distance to that patch (i.e., IBD) and then multiplied by the permeability values of all barriers between the two patches. Scenarios 2–5 considered asymmetrical probabilities to incorporate effects of one-way movement.

We further investigated how each model scenario affected survival, total population abundance, and patch population abundance. We conducted 3 Monte Carlo replicates per parameter combination to assess variance in overall population response. All simulations included a 20-year burn-in period for stabilizing population dynamics before the

respective suppression scenarios were implemented.

3. Results

3.1. Electrofishing schedules

We compared population responses for three different simulated electrofishing schedules with a baseline model of no treatment combined with IBD (Fig. 3a). The baseline model showed a stable population through time after the initial transient population dynamics (Fig. 3a; red-dashed line) allowing for independent assessment of the effect of suppression. However, results indicated little difference between electrofishing schedules in terms of long-term population recovery (Fig. 3). During treatment years there was some variation across electrofishing schedules for those patches that received treatment (Fig. 3b and d).

3.2. Three-cycle suppression scenario

To understand the cumulative effect of prolonged suppression, we simulated 3 cycles, or 30 years, of suppression activities. While some effect of consecutive cycles of suppression on population size was apparent, recovery times following suppression were not different from simulations of single-cycle suppression (Fig. 4).

3.3. Effort scenarios

During suppression there was a clear effect of electrofishing effort on population size in the suppressed patches, as greater effort resulted

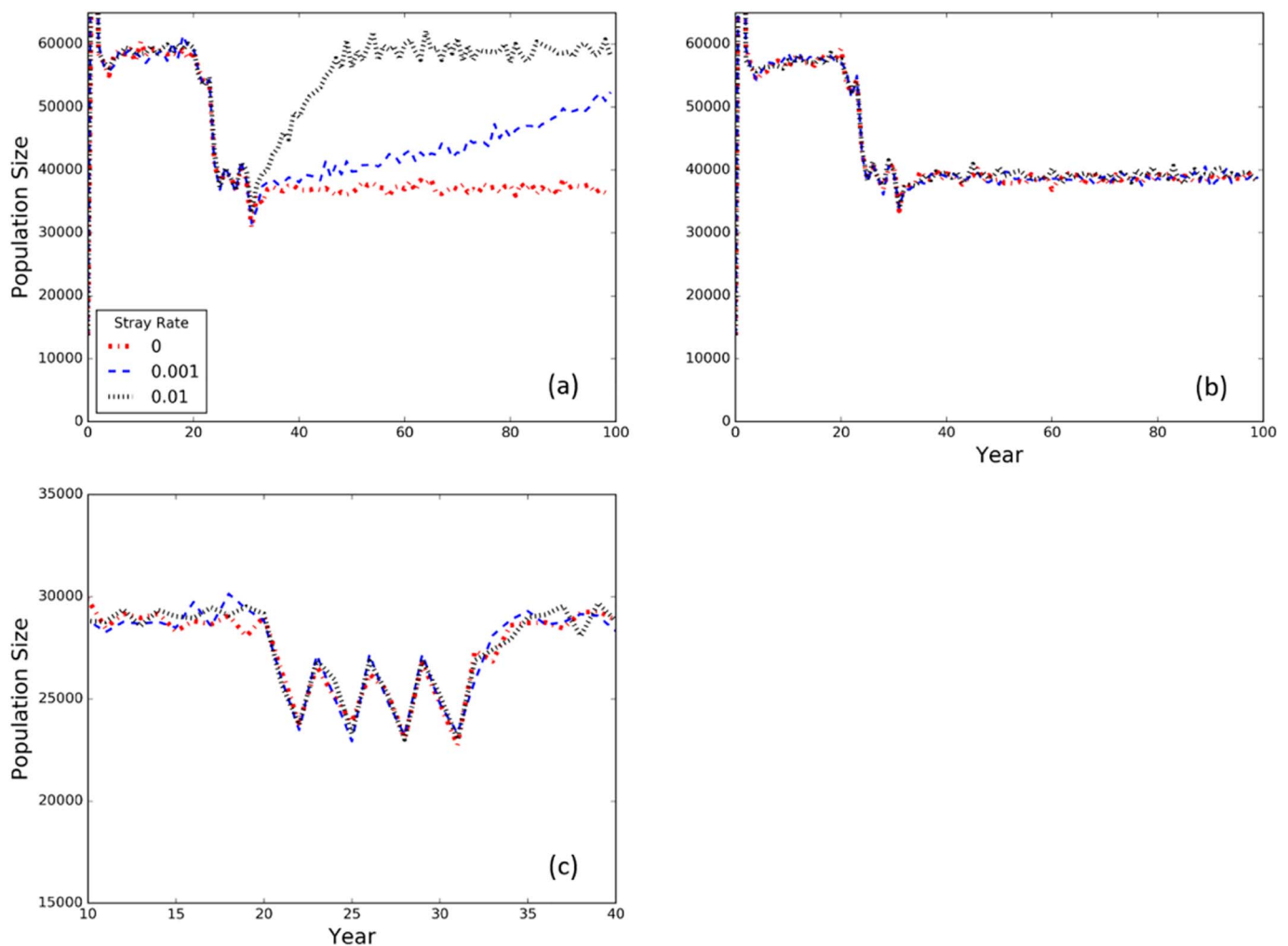


Fig. 8. Effect of stray rate on EBT control and recovery under (a) an IBD scenario and (b) an asymmetric barrier scenario. Suppression took place from years 20–31. Panel (c) displays the effect of stray rate on recovery of EBT in the subset of patches that received suppression treatment.

in lower population levels. Following suppression activities, however, recovery of populations did not differ across effort levels (Fig. 5).

3.4. Compensatory response - age 0 survival

Prior to suppression treatments, mean ($\pm$ 95% confidence interval) annual survival of age 0 fish (i.e., overwinter survival of fingerlings to yearlings) was 0.368 ± 0.0094 . During electrofishing years across all suppression scenarios, mean annual survival of age 0 fish was not significantly different from pre-suppression survival (mean = 0.381 ± 0.014 , $t = 1.25$, $p = 0.214$). In the year following suppression, however, survival increased significantly relative to pre-suppression conditions (mean = 0.612 ± 0.030 , $t = 18.17$, $p < 0.001$). These results varied by electrofishing schedule as post-suppression increases in survival were dependent on electrofishing schedule and intensity (Fig. 6).

Size structure was also affected by electrofishing treatments. In the absence of treatment, size structure remained constant throughout the simulations (Fig. 7). During years and in patches where electrofishing suppression took place, however, the proportion of total fish length represented by individuals < 110 mm increased while the proportion of total fish length represented by individuals > 180 mm decreased. The amount and pattern of fluctuation in these proportions varied according to the suppression schedule implemented (Fig. 7; Appendix A4).

3.5. Compensatory response - stray rate scenarios

We simulated three alternative stray rates to determine the strength of the relationship between fish movement and population recovery time (Fig. 8). Under the IBD scenario, recovery operated as a function of stray rate (Fig. 8a). Under the barrier scenarios, stray rate had little effect on recovery since all eradication were simulated above barriers and individuals were unable to move over barriers into eradicated reaches (Fig. 8b). When evaluating the subset of patches that received suppression treatment, suppression was not effective enough to reduce populations to the point that differences in recovery times emerged across stray rate scenarios (Fig. 8c).

3.6. Riverscape surfaces scenarios

Under the IBD scenario, populations fully recovered from eradication and suppression regardless of movement rates (Fig. 9). However, when we incorporated a riverscape surface that included barriers to fish movement, those locations that laid beyond a barrier (i.e., eradication patches) were inaccessible to immigration, and therefore EBT populations were unable to recover (Fig. 9). The suppressed areas, none of which were isolated by barriers, recovered only a few years following suppression, regardless of the riverscape surface used in the simulation (Fig. 9).

The incorporation of barriers had a strong effect on the movement and recovery of EBT in the system (Fig. 9). This is particularly true

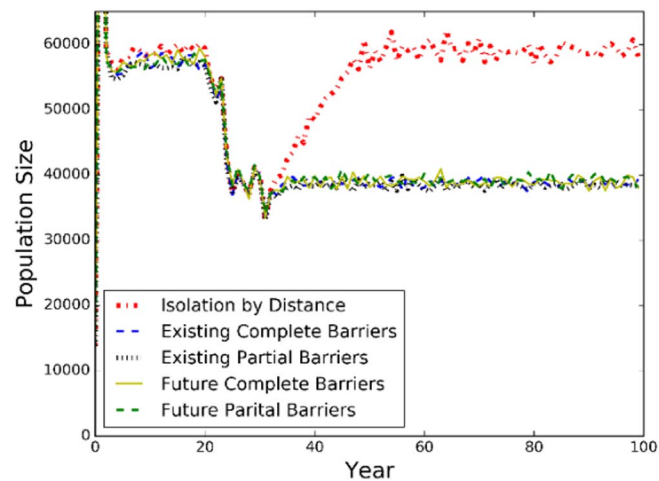


Fig. 9. Effect of riverscape scenario on EBT control and recovery over time for current and future barrier scenarios. Future scenarios incorporated several barrier removals (see Fig. 1). Complete barrier scenarios excluded the effect of barriers that allow the passage of some EBT based on assigned barrier permeability, while partial barrier scenarios incorporated these features (e.g., culverts). Stray rate was 0.01 for all scenarios presented.

because all eradicated reaches have a complete barrier to EBT at the mouth of the tributary preventing immigration into these reaches. In addition, we found virtually no effect of planned dam removals (i.e., future barrier scenarios) on EBT population size, although all simulations of planned barrier removals took place in reaches that EBT either did not previously inhabit or were eradicated. Incorporating effects of partial barriers (i.e., culverts) also had no effect on the overall population size or recovery rate of brook trout (Fig. 9).

3.7. Spatial distribution

The spatial distribution of the population of EBT on the riverscape can be more informative than knowing total population size. During simulations, total abundance was greatly reduced during the first few years of treatment due primarily to eradication in selected streams (Fig. 10a, b). Reductions in population size in suppressed reaches were significant during and immediately following a suppression cycle (Table 2; Fig. 10b), but after 10 years population levels no longer differed among treatment scenarios (Table 2; Fig. 10b, c; Video A3). Suppression reduced patch population levels to an average of 60.4% of carrying capacity for scenario 3 to 75.3% of carrying capacity for scenario 2. Standard deviations around % carrying capacity increased during suppression years, indicating that population sizes were not uniformly reduced in population size during suppression (Table 2).

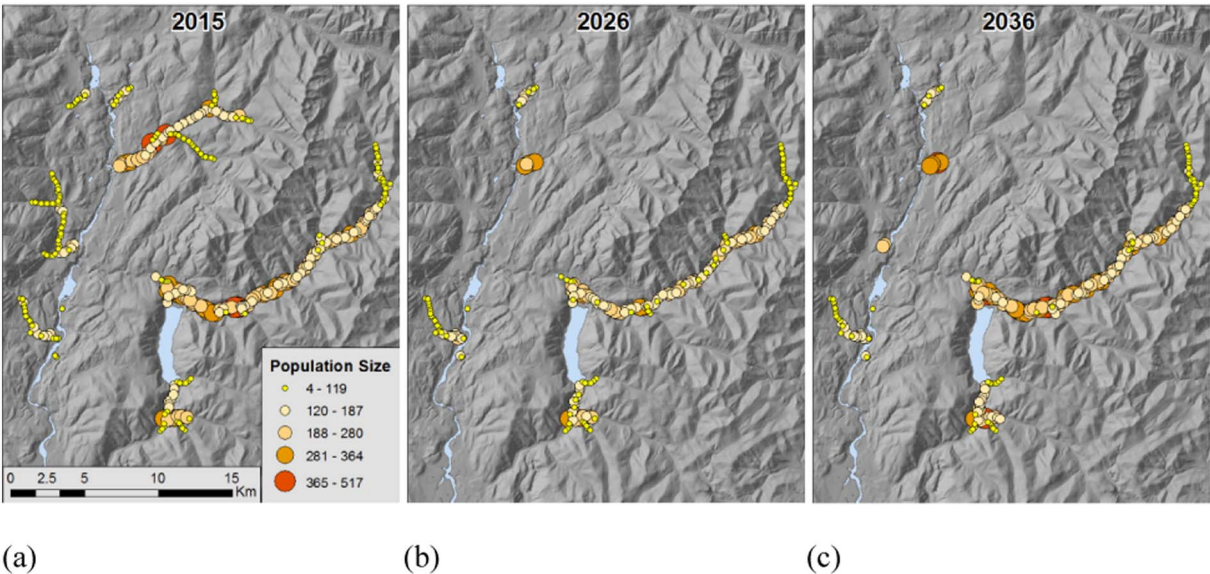


Fig. 10. Spatial distribution of individuals across the managed riverscape (a) pre-treatment, (b) during treatment, and (c) 10 years post-treatment. This simulation incorporated suppression scenario 3, a stray rate of 0.001, and the existing partial barrier surface.

Table 2
Spatially-averaged means and standard deviations of proportion of carrying capacity across patches at various points before, during, and after treatment for control of invasive eastern brook trout. Results in each row are from a single simulation run and represent population levels for the subset of patches that received electrofishing suppression treatment. The final row displays results from Analysis of Variance tests among treatments for each time step included in the table.

Treatment	Pre-treatment		During treatment		Post-treatment (1 year)		Post-treatment (10 years)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
None	0.875	0.170	0.929	0.174	0.887	0.159	0.907	0.175
Scenario 1	0.878	0.172	0.724	0.180	0.798	0.170	0.892	0.178
Scenario 2	0.906	0.162	0.753	0.181	0.762	0.198	0.903	0.174
Scenario 3	0.875	0.152	0.604	0.214	0.906	0.176	0.944	0.182
p value (F statistic)	0.242 (1.40)		< 0.001 (84.65)		< 0.001 (25.63)		0.051 (2.60)	

4. Discussion

4.1. EBT control

Our model of control of invasive EBT indicated that electrofishing was unlikely to eradicate EBT populations, though maintenance of populations at a suppressed abundance may be a viable alternative. The majority of the decline in total population size during simulations was due to eradication efforts via chemical treatments. While electrofishing suppression was able to reduce the EBT population size in suppressed patches by up to 33% (Fig. 3), full population recovery was achieved within 5 years following suppression activity in those patches. This result was robust to changes in the electrofishing schedule implemented (i.e., allocation of effort over time; Fig. 3), the length of the suppression period (Fig. 4), and suppression effort (i.e., number of electrofishing passes; Fig. 5). One possible explanation for the persistence and rapid recovery of EBT populations was low capture efficiency that varied by both location and size class. Smaller fish, which constituted the majority of individuals in the system, more efficiently avoided suppression than larger fish and contributed to population growth in subsequent years.

Historically, the response of EBT to control efforts has been mixed, with both successes and failures reported. Total eradication of EBT via electrofishing is expensive and labor-intensive, and low capture efficiencies make eradication using this method difficult in larger systems (Thompson and Rahel, 1996; Shepard et al., 2002; Meyer et al., 2006). In smaller streams (e.g., 1–3 m in width), complete eradication has been occasionally achieved (Shepard et al., 2002; Carmona-Catot et al., 2010; Buktenica et al., 2013). Factors that have contributed to the successful eradication of EBT include the size of the stream, high capture efficiency (in some cases due to manual removal of riparian vegetation), placement of barriers below the eradication zone, several consecutive years of suppression, and high labor-intensive efforts. Because attempting complete eradication often either fails or is considered infeasible (Meyer et al., 2006; Peterson et al., 2008), management plans may call for only the suppression of EBT to foster coexistence with native competitors. Peterson et al. (2004a) warned, however, that in the absence of eradication, indefinite periodic maintenance via electrofishing would be required to reduce the likelihood of the native species' decline. Our results corroborate the bulk of the literature, indicating that in large systems with inefficient capture probabilities, eradication of EBT is likely impossible. Suppression of EBT populations may be possible, but only with continued bouts of electrofishing with < 5-year gaps between sessions.

4.2. Compensatory responses

Compensatory responses to suppression played a major role in the recovery of EBT in our simulations. Examples of compensatory responses included increased overwinter survival of fingerlings (Fig. 6) and increased rates of immigration to suppressed reaches (Fig. 8). These results are again consistent with previous brook trout suppression efforts (Peterson et al., 2004a; Meyer et al., 2006; Carmona-Catot et al., 2010). Our model reproduced the increased survival of small fish as a result of the “packing algorithm” (for a detailed description of the packing algorithm, see model description in Landguth et al., 2017), which simulates density-dependent mortality by allocating resources among size classes. Changes in size structure due to electrofishing gear selectivity occurred as larger fish were disproportionately removed from the population, and more “space” became available for smaller fish (Railsback et al., 2002). This combination of factors resulted in increased survival rates for young fish following suppression. However, it is clear that this compensatory response is dependent on other model processes as well, since elevated survival did not occur uniformly across the 3 electrofishing schedules.

We also demonstrated how increased rates of straying (i.e.,

immigration) may contribute to EBT persistence (Fig. 8). In our simulations, an imposed increase in individual straying probability from 0.001 to 0.01 resulted in a nearly proportional decrease in the time needed to fully recover all eradicated streams from ~120 years to ~15 years (Fig. 8) in an IBD scenario (i.e., no barriers). Such an increase in migration rates during suppression has been cited as a primary driver of EBT recovery (Peterson and Fausch, 2003; Peterson et al., 2008). These results highlight the need for managers to consider possible compensatory responses when planning suppression and/or eradication of EBT or other aquatic invasive species (AIS), as well as the ability of SIBMs to reproduce mechanistic drivers of the invasion and persistence of exotic species.

4.3. Barriers

The use of barriers is a common means employed to prevent or delay movement of AIS into managed stream reaches and therefore isolate populations of the species of concern (e.g., Rahel et al., 2001; Novinger and Rahel, 2003). While we leave the matter of the appropriateness of such methods to others (Fausch et al., 2006), our results demonstrate how barriers might affect suppression and eradication efforts. For example, in an IBD scenario, EBT were eventually able to recolonize the entire landscape in every scenario we tested. Incorporating impenetrable barriers, however, prevented immigration to all streams eradicated with piscicides. Additional scenarios could be constructed that test the effect of faulty barriers that fail to restrict AIS movement to varying degrees. We also simulated the effect of one-way partial barriers (i.e., culverts) on EBT movement. Partial barriers were represented by cells on the landscape that had a permeability value of 1 in the downstream direction, and permeability < 1 in the upstream direction. Our results indicated that the inclusion of such barriers on the landscape might have a negligible effect on EBT population sizes or recovery times (Fig. 9), and thus only complete barriers to fish movement might be effective for the management of EBT populations. The capability to simulate partial passability over asymmetrical barriers is a significant advancement in SIBM modeling of riverscapes. In our case the partial barriers had little effect, yet in other systems a significant effect of culverts on AIS movement has been demonstrated (Whiteley et al., 2014). The application of asymmetrical partial barriers in modeling such systems could reveal insights about the effects of culvert design and remediation on landscape connectivity and population viability of both invasive and native species.

4.4. Model benefits and limitations

Incorporating spatially-explicit individual-based models into AIS management provides several benefits to managers. First, modeling can be both fiscally and ecologically cost-effective. Prior to stepping into the field, managers can evaluate alternative scenarios of disturbance, management, or climate. Strategies that yield low probabilities of success can then be excluded from consideration and managers can be more confident that field efforts will result in achievement of management goals. In our system (as parameterized), we demonstrated that electrofishing is likely an ineffective means of eradication of AIS in a large system, suggesting that managers might focus their efforts on suppression of AIS populations rather than eradication. Second, SIBMs are spatially-explicit, so that managers of AIS can identify specific locations on the landscape where problems may arise. Spatially-explicit models also allow for more realism in simulations. In our models, for example, we varied capture probability by patch to produce more realistic mortality rates that affected EBT population sizes and recovery times. We then varied factors such as suppression effort or timing to investigate the relative effect on EBT populations. Finally, our SIBM has been integrated into a plan for adaptive management (Walters, 1986), in which the model is continuously calibrated based on empirical results from ongoing field work. This feedback process enhances the

probability of project success by contributing to more accurate model predictions over time. Model results can then be used to make predictions about the response of native species to EBT control and the likelihood of restoration of the ecological community.

SIBMs that are applied to real systems often require large amounts of empirical data to produce reliable outputs. Where data are lacking, simplifying assumptions must be made about model processes, or sensitivity analyses can be conducted to explore the space around uncertain parameters. In our case, we lacked data on density-independent mortality due to mechanisms such as predation or icing events. Therefore we excluded these sources of mortality explicitly, but implicitly included them by setting patch carrying capacities relative to actual abundance estimates derived from field sampling. In addition, while we might expect this mortality to affect variability in EBT recovery rates or population sizes, we would not expect such effects to differ among treatment scenarios. We also made the assumption in our model that the probability of movement between any two patches was only affected by riverine distance and the occurrence of barriers. In reality, other factors such as slope, temperature, and food availability might affect the probability of immigration to a patch. In our model however, patch suitability was binary and was based solely on presence of EBT from field sampling. In this way, patches with unsuitable biotic or abiotic conditions were excluded from being occupied by EBT, but all suitable patches were considered equal. Finally, movement rates and distances vary widely across studies of EBT, and were unknown in this system (Riley et al., 1992; Adams et al., 2001; Curry et al., 2002). Because of this, we simulated a range of stray rates for EBT and found this parameter to be important for EBT population recovery, and might therefore be a priority for future studies.

Our EBT model represents a necessary first step in understanding the ecological and evolutionary impacts of AIS invasion and control in our study system. Results from this study will be used to evaluate the magnitudes of compensatory responses in EBT as potential loci of action for interspecific competition to be investigated in subsequent models. Such models will explicitly include multi-species dynamics, simulating feedback between EBT and Cutthroat Trout through predation and competition. We will also use this base model of EBT population dynamics to investigate evolutionary responses of AIS to control methods, increased capture efficiencies, and alternative methods of reducing EBT populations. While it is often tempting to incorporate as much realism and complexity into a simulation model as possible, it is important to start by gaining inferences from the simplest model needed to answer the proposed question. In this case, how do EBT respond to alternative methods of control? More complex questions may prompt the inclusion of additional complexity in the model once baseline mechanisms have been established. Regardless, finding the appropriate balance between complexity and inference is one of the primary challenges of simulation modeling in ecology (Loehle, 1990; Grimm and Railsback, 2005).

4.5. Utility of SIBMs

Management plans for control of invasive species rarely incorporate SIBMs into the planning and evaluation of control methods, though such management plans could benefit from this approach (e.g., Nehrbass and Winkler, 2007). Invasive species are often those that are highly successful in a non-native environment (Colautti and MacIsaac, 2004) and therefore can be especially difficult to eradicate or suppress. Managers charged with such tasks often do so with limited resources, and SIBMs provide a means whereby alternative strategies can be filtered out and prioritized, maximizing the likelihood of success without conducting trials in the field. The ability to model such systems also yields predictive capacity and fosters pro-active management as opposed to reactionary management. Once a model is developed, it can be applied to a range of potential future scenarios, whether those scenarios are due to land-use change, climate change, or management approach. In cases where endangered species may be affected by management

actions, SIBMs can help to limit damage, such as in the case of electrofishing of streams to remove sympatric invasive species. Further, the individual-based nature of SIBMs allows for the elucidation of mechanisms that drive the success of introduced species by explicitly modeling movements, processes, and interactions of individual agents. Changes to parameterizations at the individual level thus result in landscape-level emergent properties driven by the fine-scale mechanisms that occur in reality (Breckling et al., 2005). Because of these many benefits, we urge managers of threatened species and ecosystems to consider incorporating SIBMs into their invasive species management plans.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2018.01.017>.

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